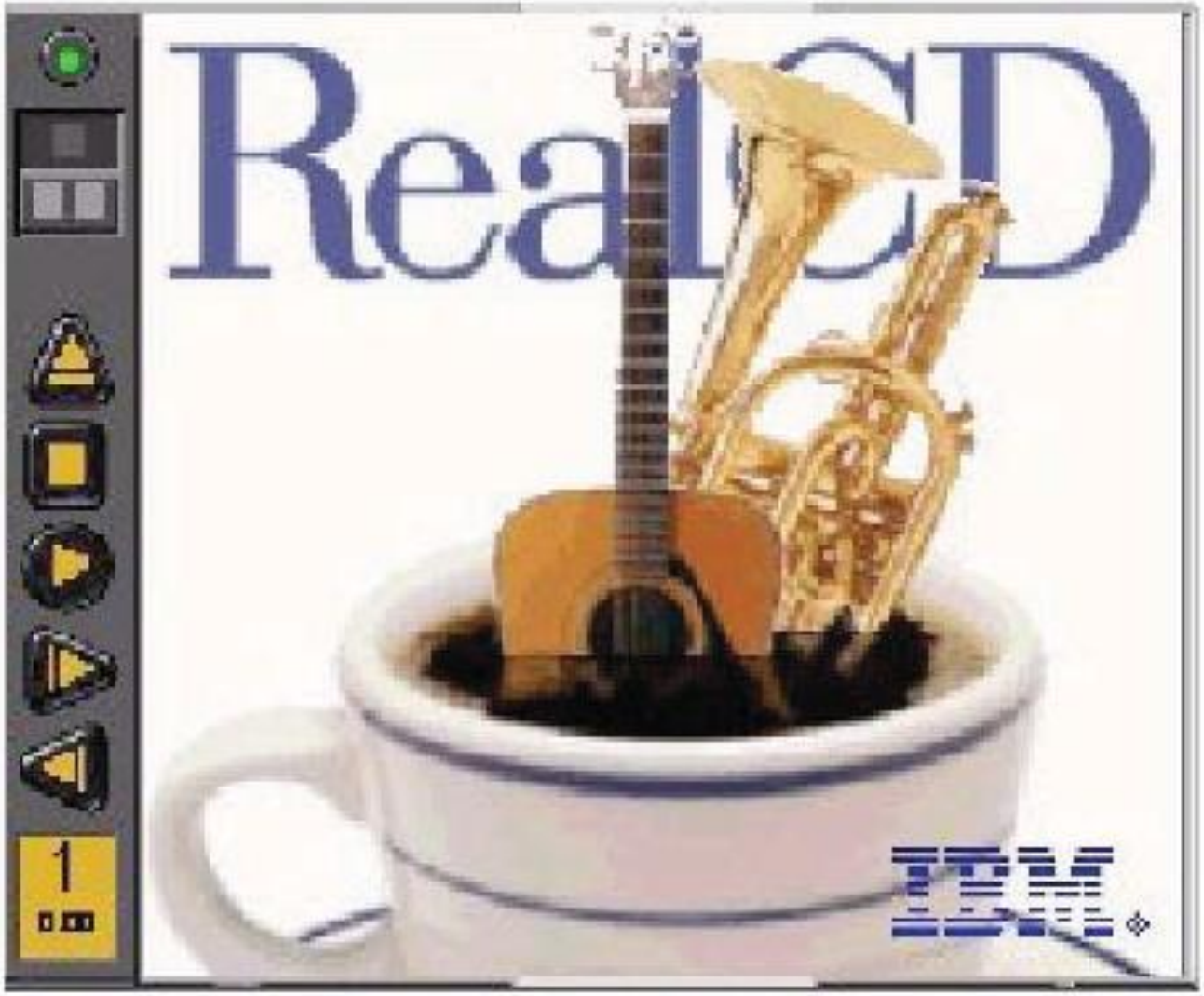
Learnability

IBM's RealCD (CD player software)



- a plastic CD case.
- *metaphor*, an analogue in the real world
- careful adherence to this metaphor produced some bad effects

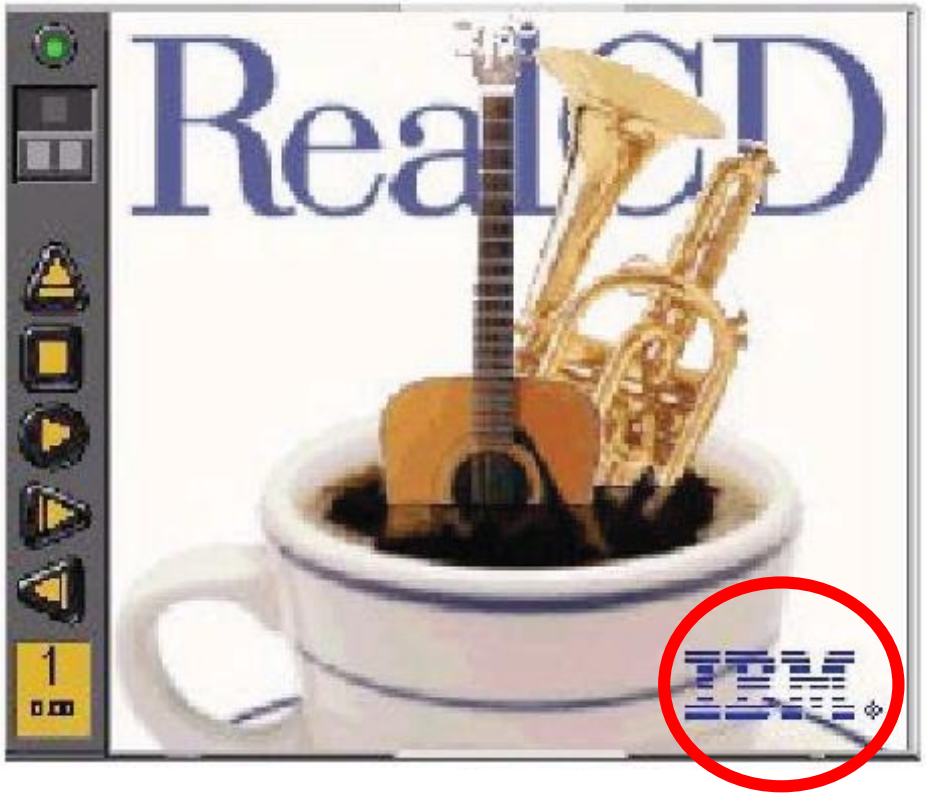
static artwork. Clicking on it does nothing.

a CD case doesn't actually play CDs

Where is this window's close box? Not more learnable with the metaphor (Consistency)

A task analysis would clearly show that adjusting the volume or picking a particular track matters more than viewing the cover art

Where is its scrollbar?



Where is the online help for this interface? the CD case must be open. You have to figure out how to do that yourself**, without help**.
With the case open, if you move the mouse over the lower right corner of the cover art, around the IBM logo, you'll see some feedback. The corner of the page will seem to peel back. Clicking on that corner will open the Help Browser

(liner notes included in a CD case)

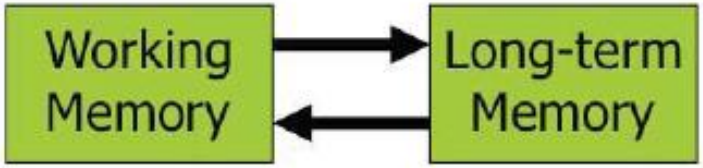
- presence of a metaphor does not at all guarantee an intuitive, easy-to-learn user interface
  - it's probably the hardest way

Today's Topics

- Human memory
- Interaction styles
- User model vs. system model
- Learnability principles & design patterns

People Don't Learn Instantly

an obvious solution to this problem would be a button that simply **executes the instructions directly**! This message is clearly a last-minute patch for a usability problem.



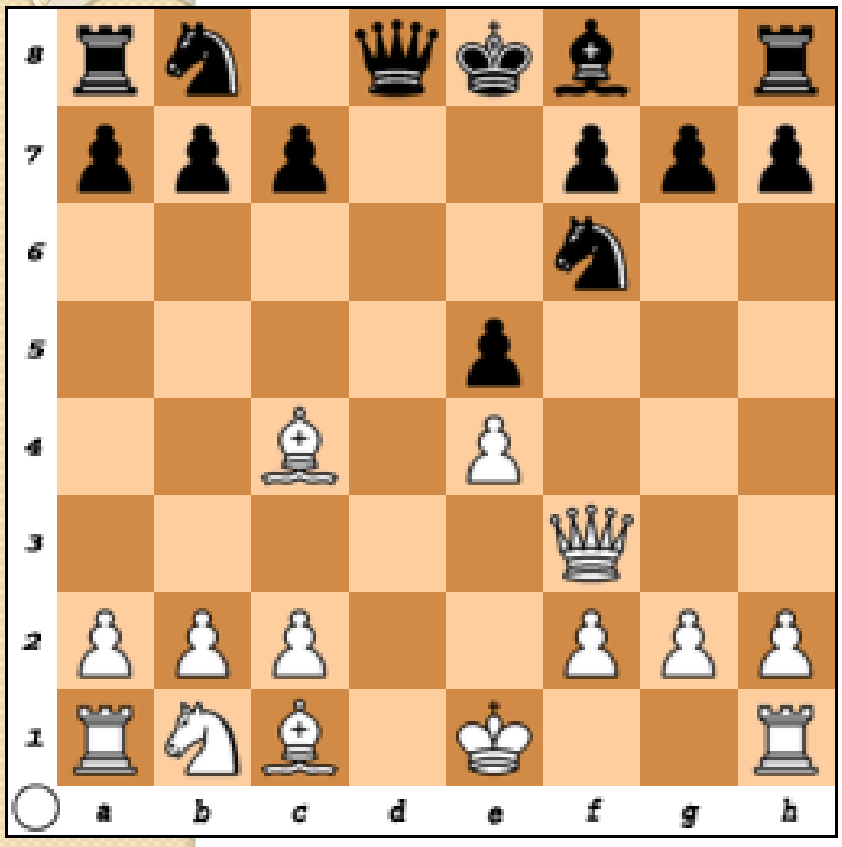
- Working memory
  - Small: 7 ± 2 "chunks"
  - Short-lived: ~10 sec
  - Maintenance rehearsal fends off decay (but costs attention)
- Long-term memory
  - Practically infinite in size and duration
  - Elaborative rehearsal transfers chunks to long-term memory

- “Chunk” is a unit of memory or perception
  - Depends both on presentation and on what you already know

Hard: M W B C R A L O A B I M B F I

Easier: MWB CRA LOA BIM BFI

Easiest: BMW RCA AOL IBM FBI



Novices and chess masters were asked to study chess board configurations and recreate them from memory

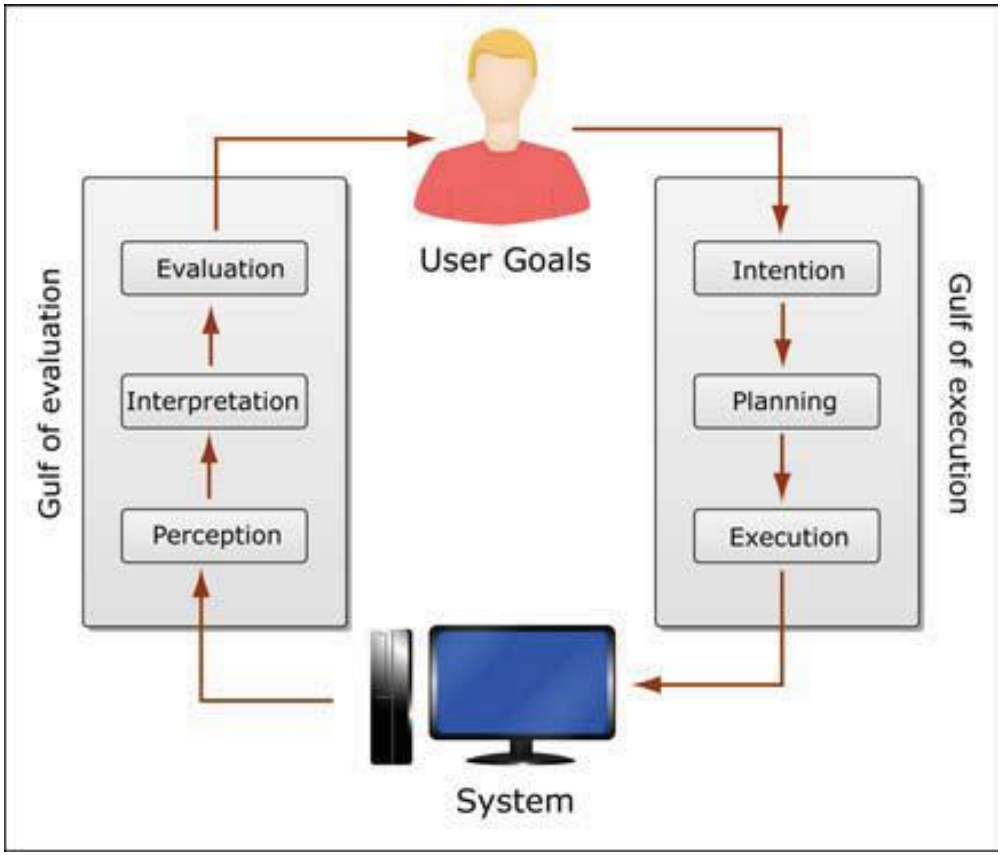
novices could only remember the positions of a few pieces.

Masters could remember entire boards, but only when the pieces were arranged in *legal* configurations

When the pieces were arranged randomly, masters were no better than novices.

- Recognition: remembering with the help of a visible cue
  - aka “Knowledge in the world”
- Recall: remembering with no help
  - aka “Knowledge in the head”

- Recognition is much easier
  - so menus are more learnable than command languages



The user has some goal in mind – let's say, using Photoshop to brighten a photo. Doing it requires constructing a plan of action and executing it. For a new user, the plan might start out with searching for a command in the Photoshop menus called something like "brightness." The user starts executing actions, and the system responds with some output. The user must then perceive, interpret, and evaluate (compare) the output against the original goal ("does this menu have a Brightness command? when I invoke it, does it do what I want?") and then either replan ("let's try the Tools menu") or change the goal ("oh well, I guess I can't brighten the image after all, I'll just add a kitten to it instead").

Interaction Styles

- Command language
- Menus & forms
- Direct manipulation

Command Language

- User types in commands in an artificial language

```
ls -l *.java
```

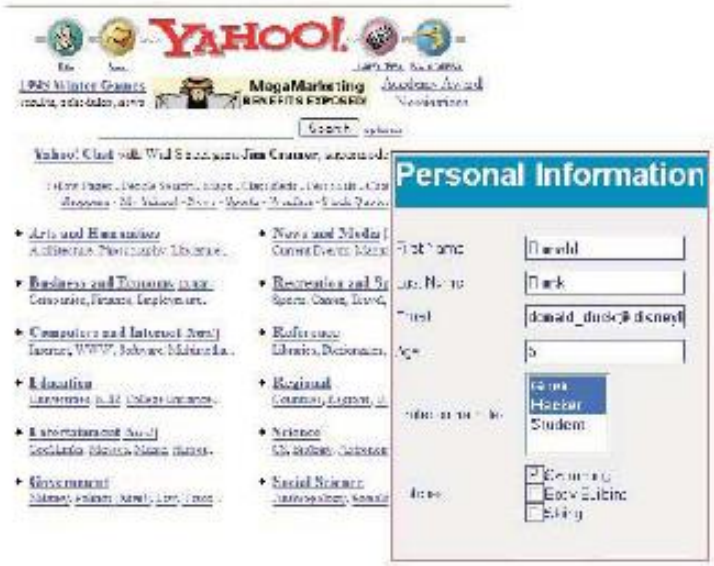
Unix shell

```
http://www.mit.edu/admissions/
```

URL

Menus and Forms

- User is prompted to choose from menus and fill in forms



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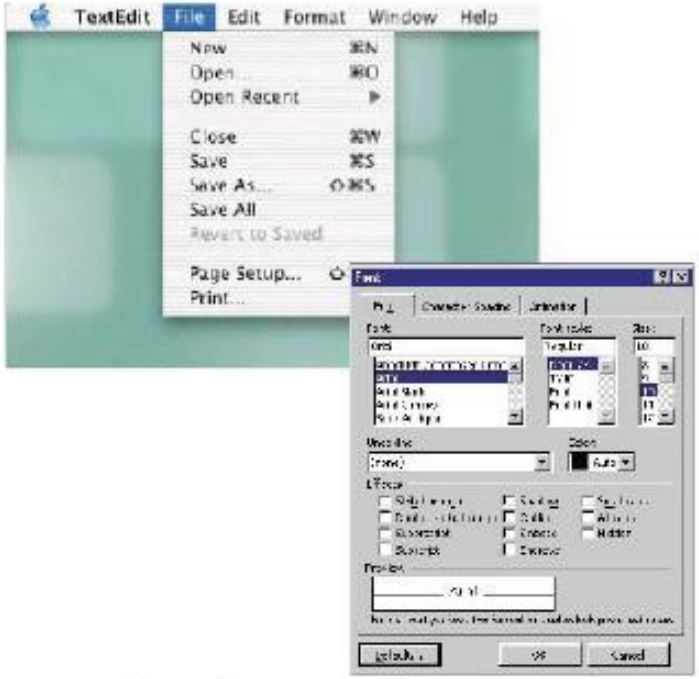

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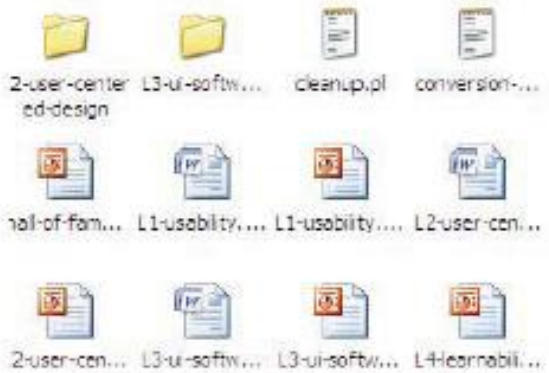
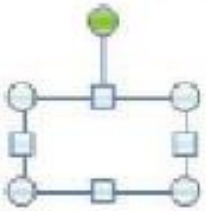
Direct Manipulation

- User interacts with visual representation of data objects
  - Continuous visual representation
  - Physical actions or labeled button presses
  - Rapid, incremental, reversible, immediately visible effects



Files & folders on desktop

Scrollbar



Selection handles

Direct manipulation

A **continuous visual representation** of the system's data objects.

user interacts with the visual representation using **physical actions** or **labeled button presses** **(**clicking on an object to select it, dragging it to move it, or dragging a selection handle to resize)

The effects of actions should be **rapid** (visible as quickly as possible), **incremental** (you can drag the scrollbar thumb a little or a lot, see incremental change), **reversible** and **immediately visible** (the user doesn't have to do anything to see the effects; a command like cp a.txt b.txt has no immediately visible effect).

Comparison of interactions

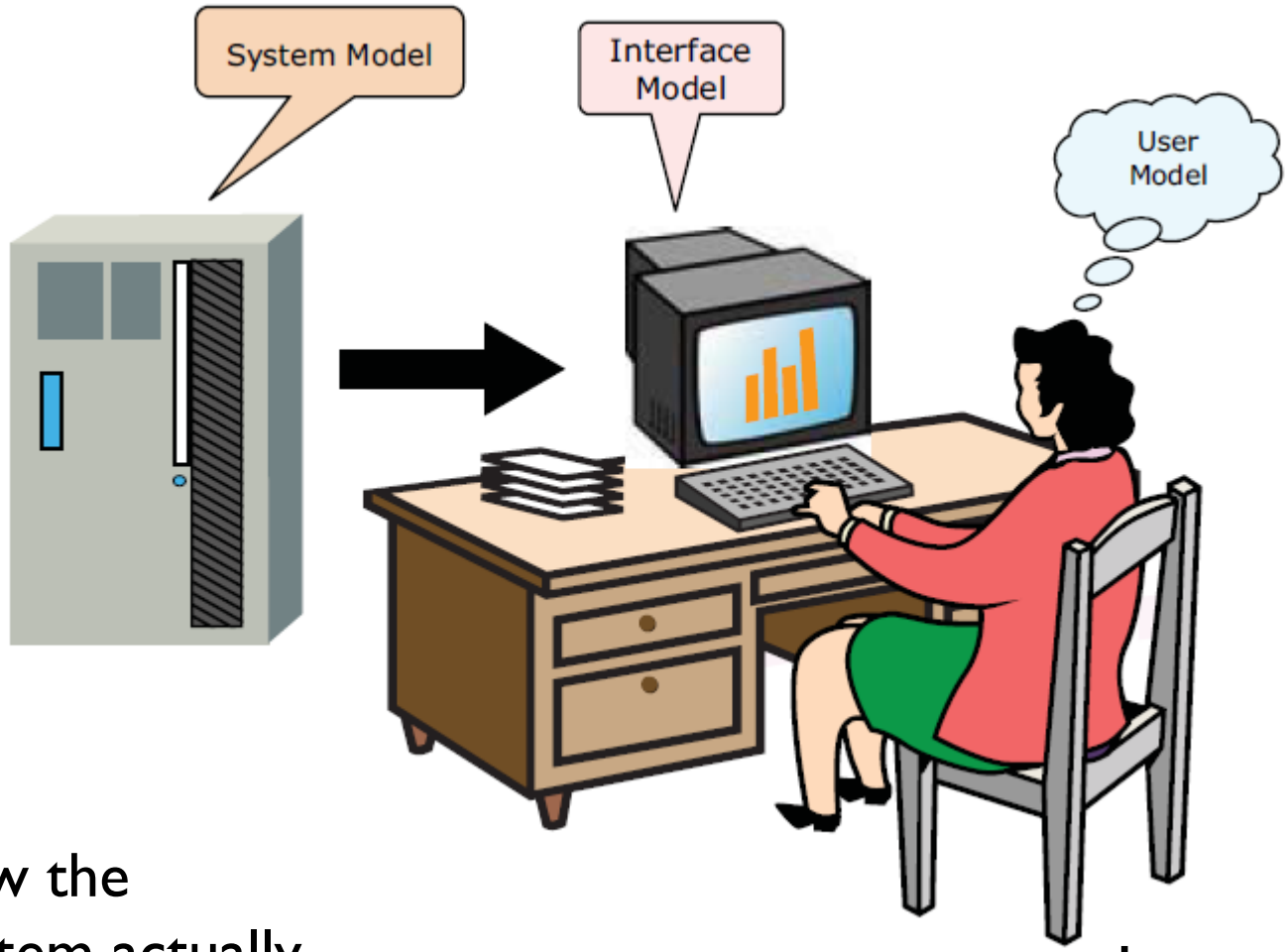
- Knowledge in the head vs. world
- Error messages
- Efficiency
- User experience
- Synchrony
- Programming difficulty
- Accessibility

Comparison of interactions

- Learnability: MF and DM are much more learnable and memorable than CL
- Error messages: CL and MF often have error messages (you didn't enter a phone number), DM rarely needs message
- Efficiency: Experts can be very efficient with CL, command histories, scripting facilitie. MF interfaces demand good shortcuts. Efficiency with DMs is possible when the DM is appropriate to the task
- User type
- Synchrony: CLs are synchronous (user types a complete command, then the system does). So MFs; you fill a web form, then you submit. DM is asynchronous: the user can point the mouse anywhere and do anything at any time
- Programming difficulty: DM is hardest to program: you have to draw, you have to handle low-level keyboard and mouse input, and you have to display feedback
- Accessibility: DM interfaces are much harder for vision-impaired users

Model

- A **way of describing how the system works**
- Photoshop and Gimp use a pixel editing model
- Illustrator use a structured graphics model
  - an image is represented by a collection of graphical objects, like lines, rectangles, circles, and text



how the system actually works

system presents to the user

how the user *thinks the system works*

The user's model may be totally wrong without affecting the user's ability to use the system

- Sometimes harmless
  - Electricity as water
- Sometimes misleading
  - Thermostat as a valve

back and previous

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Learnability Principles

- Cues that communicate the system model
  - Affordances
  - Natural mapping
  - Visibility
  - Feedback
- Consistency
  - Internal, external, metaphorical
  - Speak the user's language
  - Metaphors
  - Platform standards

Affordances

- Perceived and actual properties of a thing that determine how the thing could be used

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- Perceived vs. actual

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Natural Mapping

- Physical arrangement of controls should match arrangement of function
- Best mapping is direct, but natural mappings don't have to be direct
  - Light switches
  - Stove burners
  - Turn signals

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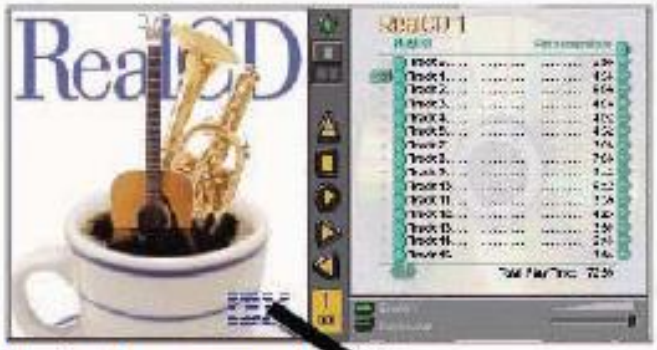
Visibility

- Relevant parts of system should be visible
  - Not usually a problem in the real world
  - But takes extra effort in computer interfaces



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mouse over

- Availability of drag & drop is often invisible

Feedback

- Actions should have immediate, visible effects
  - Push buttons
  - Scrollbars
  - Drag & drop
- Kinds of feedback
  - Visual
  - Audio
  - Haptic

Consistency

- Also called the “principle of least surprise”
  - Similar things should look and act similar
  - Different things should look different
- Kinds of consistency
  - Internal
  - External
  - Metaphorical

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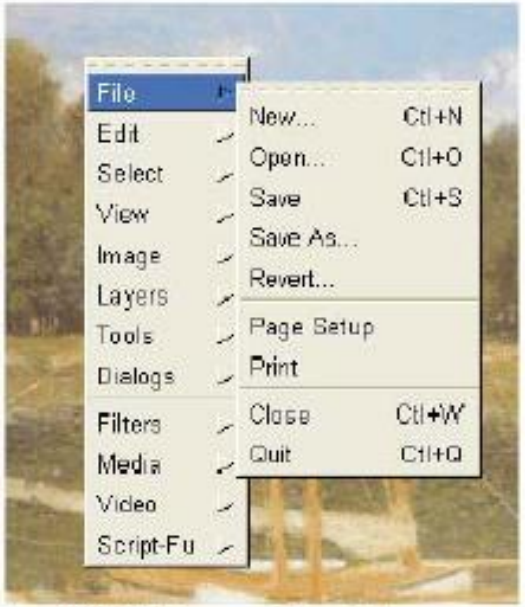
RealCD interface has problems with both metaphorical consistency (CD jewel cases don’t play; you don’t open them by pressing a button on the spine; and they don’t open as shown), and with external consistency (the player controls aren’t arranged horizontally as they’re usually seen; and the track list doesn’t use the same scrollbar that other applications do)

Consistency of Layout



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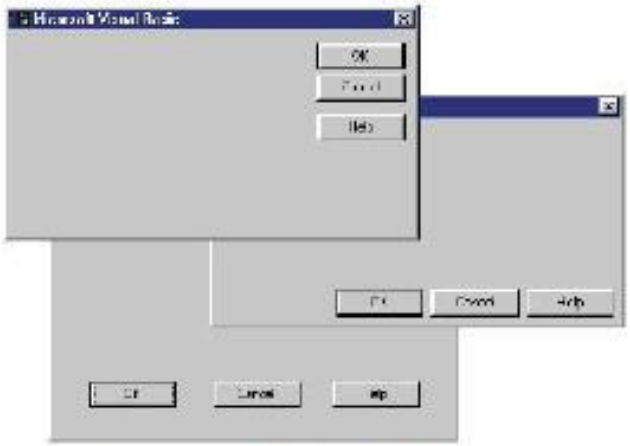

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multi-row tab widget on bottom left sacrifices consistency of layout in favor of consistency with the tabbed-notebook metaphor, not a good tradeoff. **RememberTheMilk** makes the decision in favor of layout consistency instead, keeping tabs fixed in place even it breaks the metaphor.

Consistency in Wording

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Speak the User's Language

- Use common words, not techie jargon
  - But use domain-specific terms where appropriate
- Allow aliases/synonyms in command languages

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M-I-S-M-A-T-C-H Enter

Follow Platform Standards

- Follow platform standards
  - Apple Human Interface Guidelines
  - Windows Vista User Experience Guidelines
  - GNOME Human Interface Guidelines
  - KDE User Interface Guidelines
  - Java Look & Feel Design Guidelines
- Or imitate what the popular programs do

Windows rulebook says you should have a File menu, and there should be a command called Exit on it (not Quit, not Leave, not Go Away)

pixel distances between OK and Cancel buttons on a dialog

Metaphors

- Advantages
  - Highly learnable when appropriate
  - Hooks into user's existing mental models very easily
- Dangers
  - Often hard for designers to find
  - May be deceptive
  - May be constraining
  - Metaphor is usually broken somewhere
  - Use of a metaphor doesn't excuse other bad design decisions

Desktop metaphor

Trashcan metaphor

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a word processor is like a typewriter, but you shouldn't *use* it like a typewriter. Pressing Enter every time the cursor gets close to the right margin, as a typewriter demands, would conflict with word processor's automatic word-wrapping

Deceptive metaphor: it looks like a book, but can I write in the margin? Can I rip out a page?

Strict adherence to the desktop metaphor wouldn’t scale, because documents would always be full-size like they are in the real world, and folders wouldn’t be able to have arbitrarily deep nesting

- Inconsistency is appropriate when context and task demand it
  - Arrow keys
- But if all else is (almost) equal, consistency wins
  - QWERTY vs. Dvorak
  - OK/Cancel button order

QWERTY

DVORAK

Ekran Klavyesi
Esc é " ! 1 ' 2 ^ 3 + 4 % 5 & 6 / 7 (8) 9 = 0 ? * - _ Geri Al Home PgUp
Sekme q w e r t y u I ı o p ğ ü End PgDn
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Summary

- Learnable interfaces should clearly communicate the correct mental model to the user
  - Use affordances, natural mapping, visibility
  - Consider metaphors
  - Be consistent internally, externally, metaphorically
  - Prefer knowledge in the world over knowledge in the head